

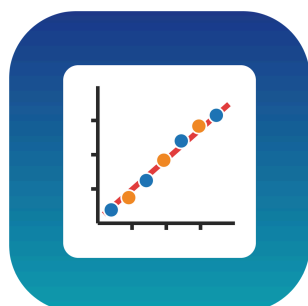


Figure Creator

A tool for quickly making publication- and presentation-ready graphs — User Manual

Version 7 · 2026-07-19

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What Figure Creator is for

Figure Creator helps you turn measurement data (CSV / TXT / Excel) into **clean, journal-ready figures quickly**, with a live preview and direct on-canvas editing. The goal is figure making — layout, styling, and export — not data analysis. You point at your columns, choose a plot style, tune the look, and export at an exact physical size and DPI.

New in v7: the whole interface is available in **English or Korean**. Switch anytime with the compact **EN/KO** button at the right end of the top toolbar. New installations start in English; your later choice is remembered.

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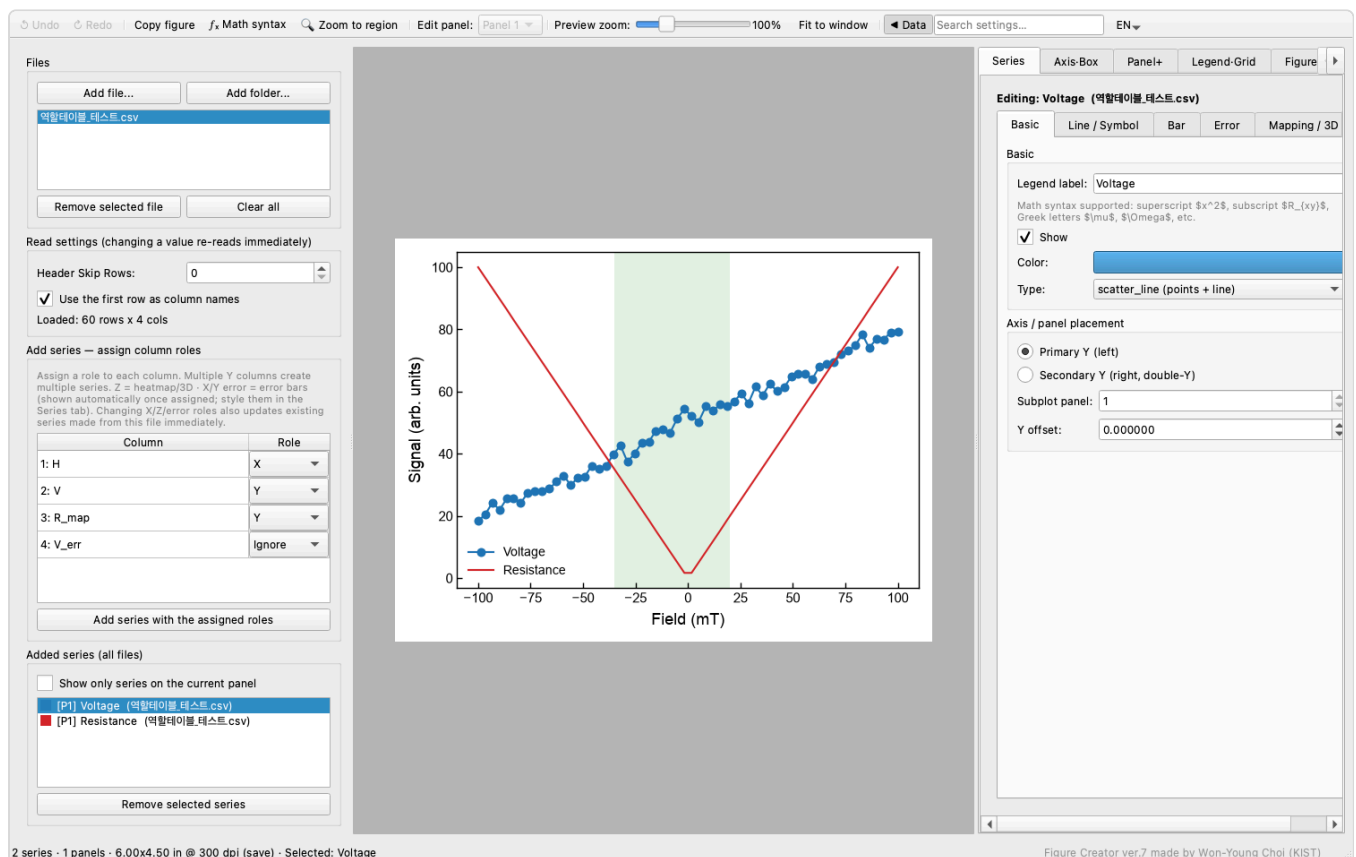
Installing and opening the app for the first time (macOS)

Important: this build is not signed with an Apple Developer ID or notarized by Apple. Only override macOS security after confirming that the download came from the creator's official distribution page. Do not override a warning that explicitly says the app contains malware or is damaged, or if the source is uncertain.

1. Download `FigureCreator-v7-macOS-AppleSilicon.zip` and double-click it to extract the archive.
2. Move **FigureCreator.app** to the Applications folder.
3. Double-click the app once. If macOS says the developer cannot be verified or Apple cannot check it for malicious software, close the alert.
4. Open **Apple menu**  → **System Settings** → **Privacy & Security**, then scroll down to the **Security** section.
5. Click **Open Anyway** for FigureCreator. This button is available for about one hour after the first launch attempt.
6. Enter your Mac login password and click **Open** again. You can launch the app normally from then on.

The current macOS build supports **Apple Silicon (M1/M2/M3/M4 family)**. It does not run on Intel Macs.

1. Screen layout



Area	Role
Left — data panel	Load files · read settings (Header Skip) · the column role table · add series / series list
Center — preview canvas	Live WYSIWYG preview. Click and drag to edit directly (see section 9)
Right — settings tabs	Series / Axis·Box / Panel+ / Legend·Grid / Figure·Preset / Notes (annotations) / Export — each with sub-tabs
Top toolbar	Undo/redo · Copy figure · f_x math help · Zoom to region · Edit-panel selector · Preview zoom · Data-panel toggle · Settings search · EN/KO

The preview-zoom slider changes **the on-screen display only**. Saved output always follows the size (inch) and export DPI set in the Figure / Presets tab.

Workspace sizing and settings search

- Drag the boundaries beside the left data panel or right settings panel to resize them. Their widths and the data panel's collapsed state are restored next time.
- Use ◀ **Data** / ▶ **Data** on the toolbar to collapse or reopen the left data panel.
- Type `colorbar`, `broken axis`, `journal`, etc. in **Search settings**, then press Enter. Figure Creator opens the matching tab, sub-tab, and control. Korean queries work too.

2. Quick start (5 steps)

1. **Add file...** — CSV / TXT / Excel supported. The number of `#` comment lines at the top of instrument files is auto-estimated (adjust manually in Read settings).
2. **Assign column roles** — in the role table, set each column to X / Y / Z / X error / Y error.
3. **"Add series with the assigned roles"** — multiple Y columns create multiple series.
4. **Refine the look** — use the tabs on the right, or click elements directly on the canvas.
5. **Export** — PNG / PDF / SVG / EPS / TIFF, or copy to the clipboard with `⌘⌥C`.

3. Loading data — the column role table

Files

Add file...

Add folder...

xrd_asdep.txt

xrd_300C.txt

역할테이블 테스트.csv

맵 격자_R_vs_H_각도.csv

각도의존성_1w_2w.csv

폴라로그진폭.csv

Remove selected file

Clear all

Read settings (changing a value re-reads immediately)

Header Skip Rows:

0

☒

 Use the first row as column names

Loaded: 60 rows x 4 cols

Add series — assign column roles

Assign a role to each column. Multiple Y columns create multiple series. Z = heatmap/3D · X/Y error = error bars (shown automatically once assigned; style them in the Series tab). Changing X/Z/error roles also updates existing series made from this file immediately.

Column	Role
1: H	X
2: V	Y
3: R_map	Z
4: V_err	Y error

Add series with the assigned roles

Added series (all files)

☐ Show only series on the current panel

[P1] as-deposited (xrd_asdep.txt)

[P1] 300 °C anneal (xrd_300C.txt)

Remove selected series

You assign a **role** to each column, and that role **is** the column's meaning. Roles are remembered per file.

Role	Meaning
X	Horizontal axis (angle on a polar panel)
Y	Vertical axis — each Y column becomes its own series
Z	Color value for a heatmap / z coordinate for a 3D plot
X error / Y error	Error bars — shown automatically once assigned (style them in Series > Error)

Retroactive: changing the X / Z / error roles updates the series already made from this file immediately. Only the Y role applies when you next add series.

Turn on **"Show only series on the current panel"** above the series list to tidy things up when you have many panels.

4. Series styling

Editing: as-deposited (xrd_asdep.txt)

Basic | Line / Symbol | Bar | Error | Mapping / 3D | Fit

Basic

Legend label:

Math syntax supported: superscript x^2 , subscript R_{xy} , Greek letters μ , Ω , etc.

☒ Show

Color:

Type:

Axis / panel placement

☒ Primary Y (left)

☐ Secondary Y (right, double-Y)

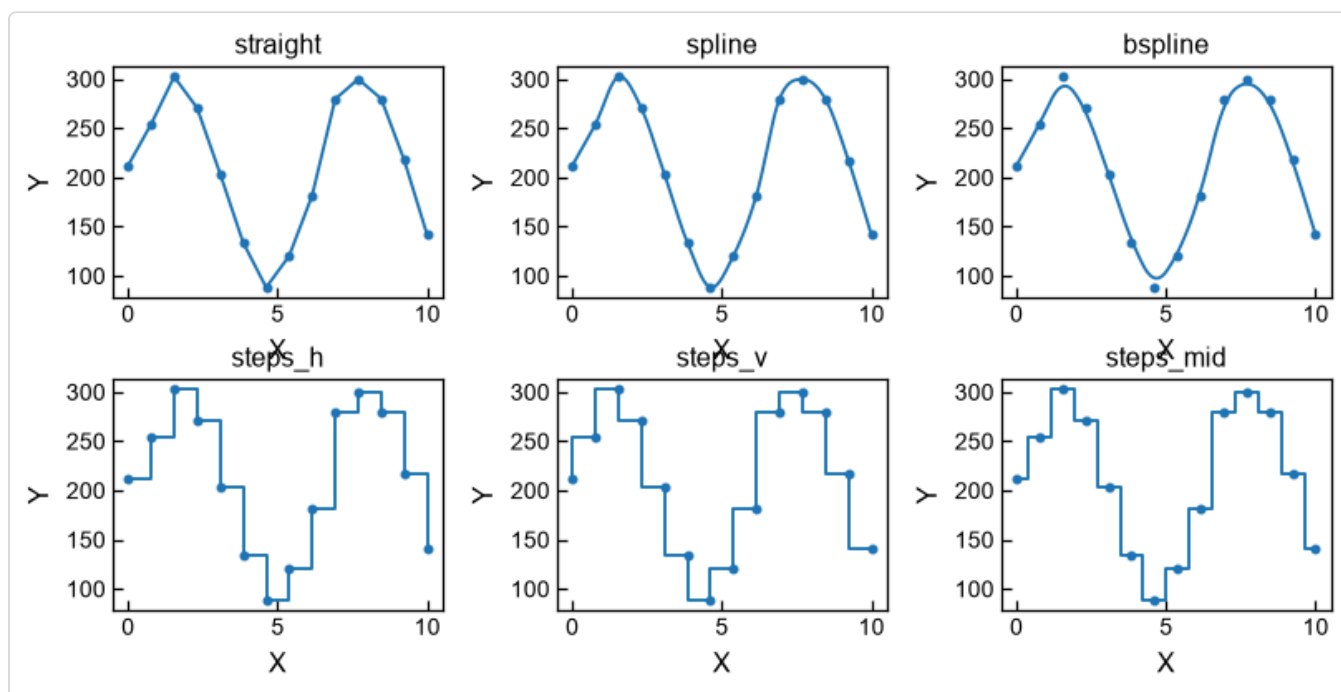
Subplot panel:

Y offset:

Applies to the one selected series. Sub-tabs:

Sub-tab	Contents
Basic	Legend label (math allowed) · show on/off · color · type (scatter / line / bar / heatmap / surface) · Y axis (primary / secondary) · which panel · Y offset
Line / Symbol	Line width/style · connect mode (straight / spline / B-spline / 3 step styles) · fill under line · marker shape/size/open/edge
Bar	Bar width ratio · opacity · edge (multiple bar series are auto-grouped)
Error	Cap size · line width · show every N · color (assign columns on the left)
Mapping / 3D	11 colormaps · manual color range (min/max) · colorbar and its style (number/label size, thickness, gap, outline, ticks)
Fit	Optional linear guide line ($ax+b$ / ax) with equation and R^2 , extrapolation, and $x=0$ extension — a convenience for illustration, not an analysis tool

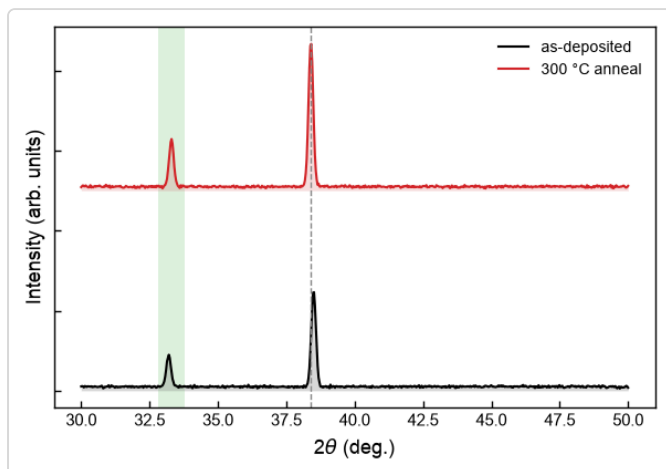
Line connect modes



A spline passes through every point; a B-spline is the smoothest and only approximates the points. Both are safe for data where x reverses, such as hysteresis loops.

5. Guide by plot type

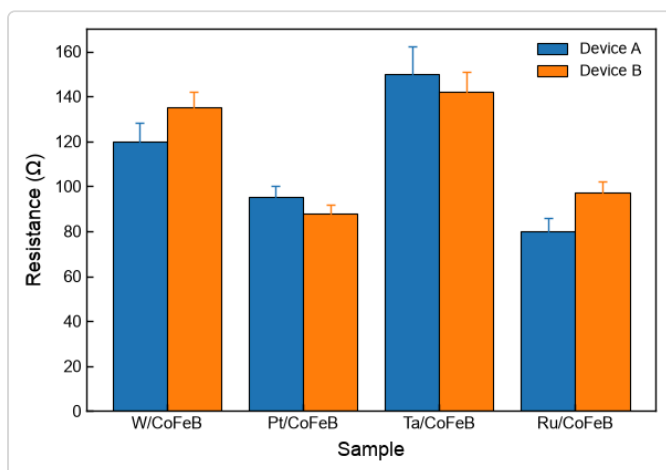
5.1 XRD-style offset stack



Axis · Box > Advanced > **Offset stack**: the Nth series gets $N \times$ step added automatically, so curves stack upward. It adds to each series' own Y offset in the Series tab.

Turn off "Show tick numbers" to make an arb.-units figure. This example combines reference lines (gray dashed), a shaded span (green), and fill-under.

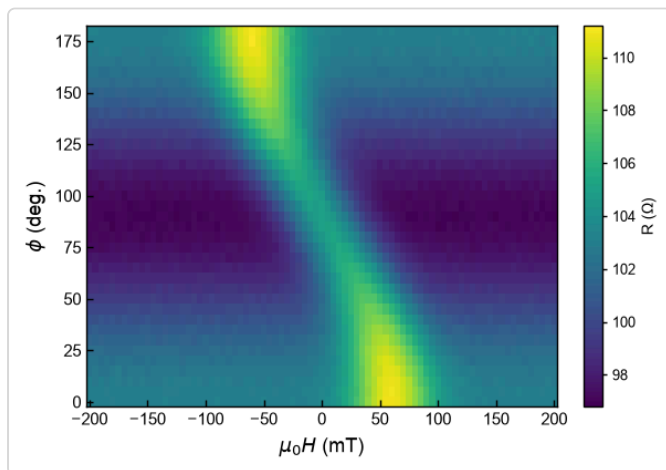
5.2 Bar charts



Bar series on the same panel are automatically placed side by side (grouped) per x position — series from different files are aligned by x value (number) or category name (text).

Assign a Y error column and the error bars appear too.

5.3 2D mapping (heatmap)

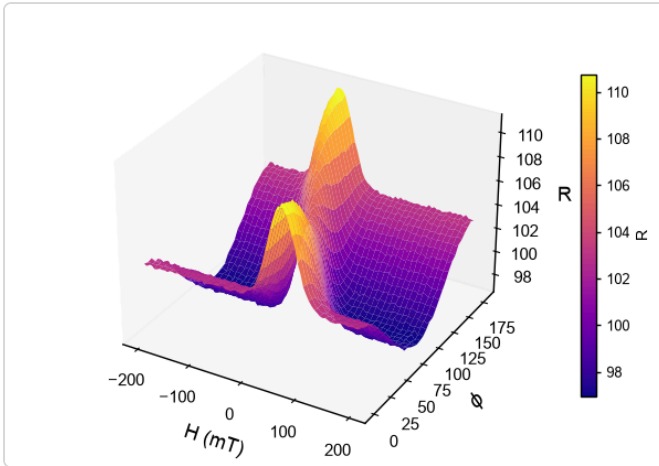


On a normal panel set the type to **heatmap**; a Z-role column is required. The (X, Y, Z) columns become a colormap.

Regular grids (like an instrument sweep) render as an exact pixel map; irregular scatter is drawn by automatic triangulation.

Click the colorbar to jump straight to Series > Mapping / 3D.

5.4 3D plots (line · points · surface)



In Panel+ > Type / Layout, set the panel type to **3d**.
A Z-role column is required.

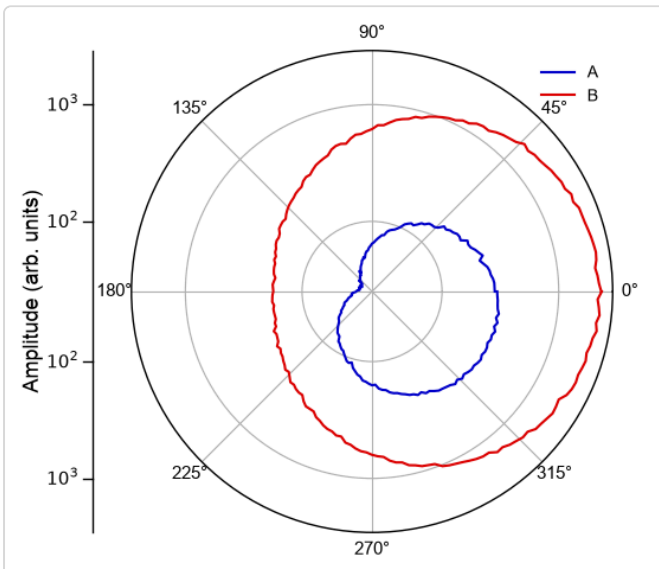
- scatter / line / scatter_line → 3D curves and points
- surface → 3D surface (colormap)

Drag with the mouse = rotate the view (release to save the angle, synced with the Panel+ spin boxes).

Option +drag = move the panel.

The Z-axis label and range are in Axis · Box > Y2 · Z.
If the colorbar overlaps the Z-axis label, increase "Colorbar gap".

5.5 Angular / polar plots

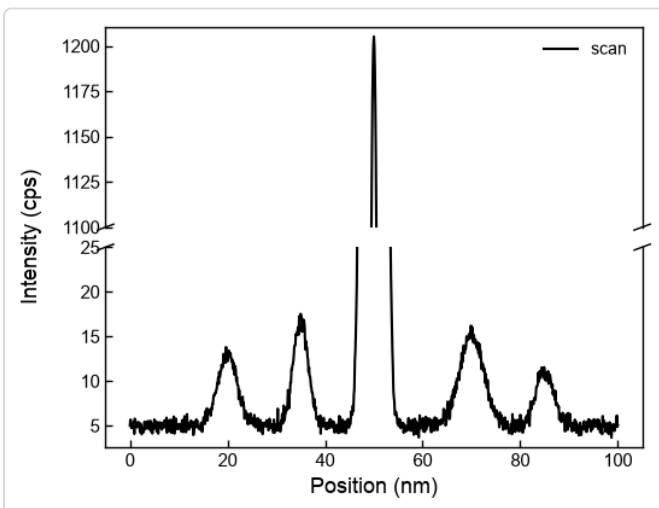


Set the panel type to **polar**: X column = angle (choose degrees or radians), Y column = radius.

Two r-tick positions (Panel+ tab): inside (default), or an **outside vertical axis** — the axis line, ticks, and numbers run vertically outside the circle, with "Outside axis gap" controlling the margin.

A logarithmic r axis (Axis · Box > Y-axis Log scale), the angular tick spacing (°), and the r-tick spacing (Axis · Box > Y-axis tick spacing) are all adjustable.

5.6 Broken Y axis



Axis · Box > Advanced > **Broken Y axis**: cuts out the given range and joins the lower and upper segments with a // mark — for showing a small signal and a huge peak in one figure.

A manual Y range (min/max) is applied as the outer bounds.

Rectangular panels only — cannot be combined with double-Y, bar, or heatmap.

6. Axis & box settings

Editing: Panel 1 (switch panels from the top toolbar)

X axis Y axis Y2-Z Box Advanced

Y axis (Primary)

Label: Intensity (arb. units)

☒ Show label ☐ Show tick numbers

Label font size: 13.00

☐ Bold

Tick font size: 11.00

Label gap: 6.00

Tick number gap: 3.50

☐ Log scale

Tick notation: auto (default, shows a 1e-6-style factor) ▼

Unit: e.g. V, A, Ω (used only in engineering mo...)

Scale order 10[^]: 0

For scaled mode only. e.g. -6 → 1e-6 shows as '1'. Add the unit like (μ) into the label yourself.

Tick spacing: Leave blank for auto (e.g. 0.5, 100, 5e-7)

☒ Auto range

min 0 max 0

Sub-tabs: **X-axis** / **Y-axis** / **Y2 · Z** / **Box** / **Advanced**. For each axis:

- Label (math allowed) + **Show label** / **Show tick numbers on-off** (for arb.-units figures)
- Font size, Log scale, auto/manual range — a manual range **accepts exponent notation (2.5e-6)** with no digit limit

- Tick notation: auto / plain / scientific / engineering (μ , m, k...) / scaled (divide by 10^n)
- Tick spacing (major tick step)

Box: spine and major/minor ticks on/off for each of the four sides, tick direction (in/out), length, width, and spine width.

Advanced: panel title · broken Y axis · offset stack.

On the canvas, **clicking an axis label or a tick number** jumps straight to the matching field (including the 3D Z axis and polar outside ticks).

7. Multiple panels and layout

Size / StyleLayoutSave

Size / Resolution

Width (inch):6.00

Height (inch):4.20

Export DPI:300

= 15.24 cm x 10.67 cm

Journal spec:Choose a journal preset...

Recommended max height: most journals 9.0~9.5 in. Resolution: 300 dpi (min) / 600 dpi (preferred).
Nature 3.50/7.20 · APS 3.37/6.97 · ACS 3.25/7.00 · Wiley 3.50/7.20 · Elsevier 3.54/6.69 (single/double column, inch)

Font:Arial

Color palette:Default (matplotlib tab10)

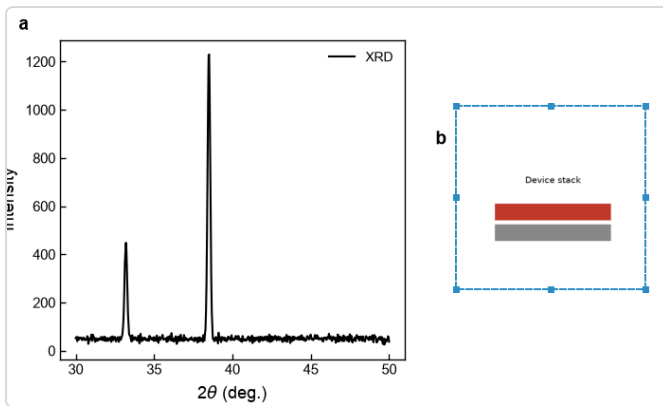
Recolor existing series with the palette

Grid mode

- Set rows × cols and the row/column gaps
- **Drag the boundary between panels** to change the row/column size ratio (1:1 magnet at equal size)

- Share X/Y axis + **Join panels**: zero gap, labels on the outside only, duplicate boundary ticks tidied automatically

Free layout



Place each panel's position and size freely — e.g. a large graph on the left plus an image on the right.

- Drag a panel = move; edge/corner handles = resize (**Shift** = keep ratio)
- Snap magnet: align to other panels' edges · match size · figure borders
- Arrow keys = nudge (**Shift** = larger, **Option** = finer)
- Right-click = copy/paste style · duplicate · delete (or **Delete**)

8. Panel labels and image panels (Panel+ tab)

Editing: Panel 1 (switch panels from the top toolbar)

Type / Layout Label / Image

Panel type

Type: Normal (Cartesian) ▼

☒ Angle column is in degrees (°) (unchecked = radians)

r tick position: Inside (default — along 22.5°) ▼

Outside axis gap: 0.08

Angular tick spacing (°): 45

Radial tick spacing and range are set from the Y-axis 'Tick spacing' and 'Auto range' in the Axis / Box tab.

3D view elevation (elev): 30

3D view azimuth (azim): -60

polar: X col = angle, Y col = radius. 3D: assign a Z column in the Series tab — scatter/line become 3D curves, surface becomes a 3D surface. heatmap stays on a normal panel.

Panel position / size (free-layout mode only)

Enabled when you switch the layout to 'free' in the Figure / Presets tab. Values are figure fractions (0~1): x-y is the panel's bottom-left corner.

X position: 0.1

Y position: 0.1

Width: 0.8

Height: 0.8

- **Panel type:** normal / polar / 3d + per-type options (angle unit, r-tick position/spacing, 3D view)
- **Panel label:** paper-style a, (b), A... free text, four positions (left/right × inside/outside) + fine-tuning
- **External image panel:** place a PNG / JPG / SVG in a panel (schematic, TEM, etc.). fit (keep ratio) / fill / manual (zoom — the occupied area grows)

- **Position / size:** numeric panel rect entry in free-layout mode

9. Interactive canvas

Action	Result
Click a panel / click empty space	Select the panel (blue dashed outline + handles) / deselect
Click a curve, marker, or bar	Select that series + open the Series tab
Click an axis label or tick number	Jump to the matching Axis · Box sub-tab and field (X/Y/Y2/Z and polar)
Click the colorbar	Series > Mapping / 3D (colorbar settings)
Click + drag the legend	Open the Legend · Grid tab + move it
Drag an annotation	Move the annotation
 /Ctrl + wheel · trackpad pinch	Zoom the preview
Toggle  Zoom to region then drag ·  /Ctrl + drag	Magnifier (box zoom): the selected rectangle becomes the axis range (rectangular panels, log axes supported)
Double-click (magnifier mode)	Restore auto axis range
Drag a panel / drag a handle	Move / resize (free layout,  = ratio)
Drag inside a 3D panel	Rotate the view ( + drag = move the panel)
Drag a grid boundary	Change the row/column size ratio
Right-click a panel	Copy/paste style · duplicate · delete · reset ratios
Arrow keys / 	Nudge / delete a panel (free layout)

10. Annotations · reference lines · spans (Notes tab)

Editing: Panel 1 (switch panels from the top toolbar)

Text annotations Ref. lines / spans

Reference lines / spans

Draw a vertical/horizontal reference line at a point (e.g. a peak position) or shade the band between two positions. Positions are in data coordinates (polar panel: a vertical line = angle).

vertical Line @ 38.4
vertical span @ 32.8~33.8

Add line Add span Remove

Orientation: Vertical (x position) ▾

Position: 0 ▴ ▾

Position 2 (span): 0 ▴ ▾

Line style: dashed ▾

Line width: 0.10 ▴ ▾

Span opacity: 0.05 ▴ ▾

☐ Gradient (fades from position to position 2)

Color:

- **Text annotations:** text (math allowed) pinned to data coordinates + an optional arrow. Draggable on the canvas.
- **Reference lines:** vertical (x position) / horizontal (y position) marker lines — e.g. to mark a peak position. Color, line style, width.

- **Spans:** a shaded band between two positions (opacity, with a **gradient** option — fading from position to position 2)
- Polar panel: a vertical line = an angular ray, a horizontal line = a radius circle

11. Math (LaTeX) input

Tip: in any label / legend / annotation / panel title, only the part wrapped in `$...$` is treated as math. Use braces `{}` to group several characters: `$10^{(-6)}$` (OK) vs `$10^-6$` (wrong). After a backslash command, add a space or braces: `$\mu_0$`, `$\theta$`.

Input (copy and use)	Rendered result
Superscript <code>\$x^2\$</code> <code>\$10^{-6}\$</code> <code>\$e^{-t/ \tau}\$</code>	x^2 10^{-6} $e^{-t/\tau}$
Subscript <code>\$R_{xy}\$</code> <code>\$H_{ext}\$</code> <code>\$M_s\$</code>	R_{xy} H_{ext} M_s
Super + subscript <code>\$V_{xy}^{2\omega}\$</code>	$V_{xy}^{2\omega}$
Greek letters <code>\$\alpha\$</code> <code>\$\beta\$</code> <code>\$\gamma\$</code> <code>\$\theta\$</code> <code>\$\phi\$</code> <code>\$\mu\$</code> <code>\$\rho\$</code> <code>\$\tau\$</code> <code>\$\omega\$</code> <code>\$\Delta\$</code> <code>\$\tau\$</code> <code>\$\omega\$</code> <code>\$\Omega\$</code> <code>\$\Delta\$</code>	α β γ θ ϕ μ ρ τ ω Δ
Common combinations <code>\$\mu_0\$</code> <code>H\$</code> (mT) <code>\$2\theta\$</code> (deg.) <code>\$\phi_H\$</code> (deg.) <code>\$\phi_H\$</code> (deg.)	$\mu_0 H$ (mT) 2θ (deg.) ϕ_H (deg.)
Fraction <code>\$\frac{V_{xy}}{R_0}\$</code> <code>\$\frac{\Delta R}{R_0}\$</code>	$\frac{V_{xy}}{R_0}$ $\frac{\Delta R}{R_0}$
Square root <code>\$\sqrt{H^2 + H_k^2}\$</code>	$\sqrt{H^2 + H_k^2}$

Close

In any label, legend, annotation, or panel title, the part wrapped in `$...$` is rendered as math.

The **f_x Math syntax** button on the toolbar opens a cheat sheet of 12 common patterns shown with live rendering (the code on the left is copyable).

e.g. `$\mu_0$ H$ (mT)`, `$2\theta$ (deg.)`,
`$R_{xy}$`, `$V_{xy}^{2\omega}$`

12. Journal specs · color palettes (Figure / Presets > Size / Style)

Journal	Single column	Double column
Nature / Nature family	3.50" (89 mm)	7.20" (183 mm)
APS (PRL, PRB, PRM...)	3.37" (86 mm)	6.97" (177 mm)
ACS (Nano Lett. ...)	3.25" (83 mm)	7.00" (178 mm)
Wiley (Adv. Mater. ...)	3.50" (89 mm)	7.20" (183 mm)
Elsevier (JMMM ...)	3.54" (90 mm)	6.69" (170 mm)

Choosing one from the combo applies the width immediately, and a **cm conversion** is shown live below the size spin boxes. Max height is typically 9.0–9.5", and resolution 300 dpi (min) / 600 dpi (preferred).

Color palettes: Default (matplotlib) / Colorblind-safe (Okabe-Ito) / Origin style / Muted / High contrast — applied to new series' colors, and "Recolor existing series with the palette" reassigns them all at once.

13. Saving presets and projects (Figure / Presets > Save)

	Style preset	Project
Contents	The whole figure style (no data)	Style + series + file paths
Use	Reuse a look, e.g. "style for a Nature single column"	Reopen the entire piece of work later
Format	JSON — compatible with earlier-version (v1–v6) files	

A project stores the **file paths**, not the data itself. If you move the original data files, you'll get a "could not find" notice when reopening.

14. Exporting

- Formats: PNG / PDF / SVG / EPS / TIFF — several at once
- Saved size = the inch size from the Figure tab exactly (no cropping), resolution = export DPI
- **Copy figure** (  C): a PNG at the export DPI to the clipboard — paste straight into slides

15. Shortcuts / troubleshooting

Shortcuts

 Z /  ⌘ Z	Undo / redo (a drag or a run of edits is grouped into one step)
 ⌘ C	Copy the figure to the clipboard
 K / Ctrl+K	Focus Search settings
 ⌘ B / Ctrl+Shift+B	Collapse / expand the left data panel
Arrow keys ( Shift /  Option)	Move the selected panel — larger / finer (free layout)
Delete	Delete the selected panel (free layout)

Common issues

Symptom	Fix
File shows ⚠ and won't load	Set Header Skip Rows to the number of comment lines at the top of the file
Text looks broken in the figure	The figure font (Arial, etc.) has no glyphs for that script — use Latin text in labels
heatmap/3D says "needs a Z column"	Assign a Z-role column in the role table on the left
Colorbar overlaps the Z-axis label	Increase Series > Mapping / 3D > Colorbar gap
A 3D panel won't move	Dragging inside rotates it — use <code>Option</code> +drag to move, edges to resize
Error bars don't show	Check that an X/Y error column is assigned in the role table on the left
Undo a magnified range	Double-click the plot in magnifier mode, or check "Auto range" in the Axis · Box tab. <code>⌘Z</code> works too
The left data panel is missing	Click ► Data on the toolbar or press <code>⌘⇧B</code>
Unidentified developer warning	Follow "Installing and opening the app for the first time" and use Open Anyway in Privacy & Security